

Sex and the City:

Spatial Structural Changes and the Marriage Market

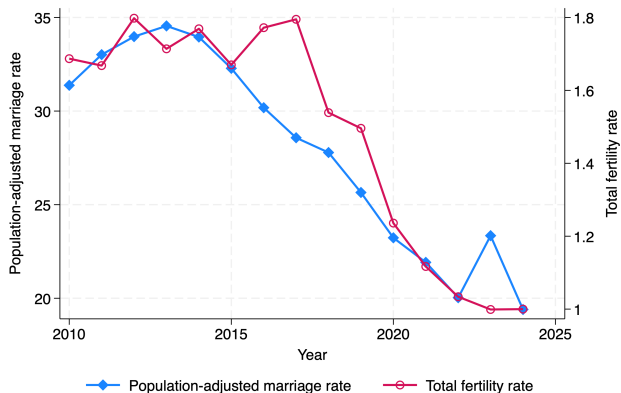
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Marriage and Spatial Sorting: Motivation

- ▶ Marriage and fertility rates are declining globally;
- ▶ It is not only for developed economies, but also for developing countries.



Marriage and Spatial Sorting: Motivation

- Spatial disparity in marriage $\Rightarrow$ aggregate marriage rate $\downarrow$

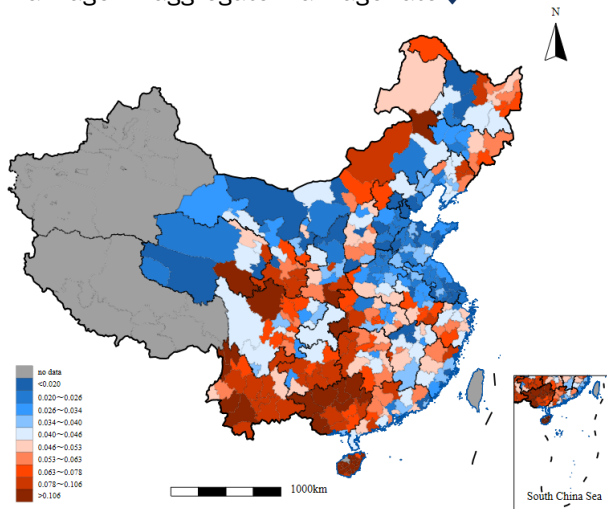


Figure Single Rate Gap (Male-Female) for People aged 30 to 45 (living location)

Marriage and Spatial Sorting

- ▶ **Spatial mismatch** in marriage market:

There are more single

- * (high-skill) women in developed/urban areas;
- * (low-skill) men in less developed/rural areas.

- ▶ Previous studies investigate these two issues separately (Ong et al., 2020; Edlund et al., 2013)

- ▶ What if they are two sides of the same coin from a spatial equilibrium perspective?

- * Both jobs and marriage are local;
- * People (by gender & skill) make migration decisions incorporating considerations of both.

Goal: Understand & quantify the sources of observed patterns in the spatial marriage market.

This Paper (Empirical)

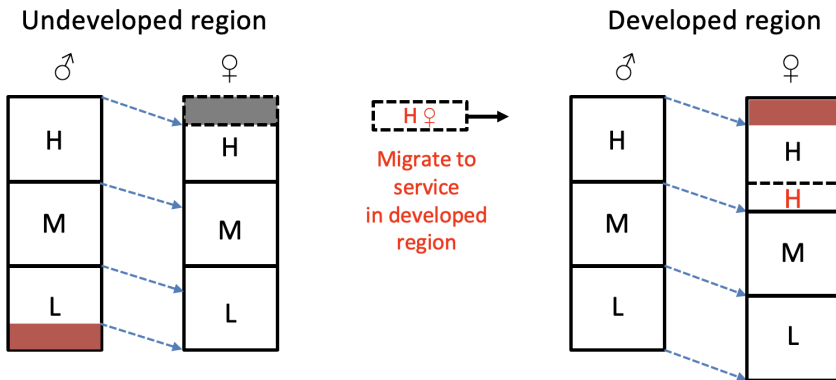
- ▶ Three stylized facts on China's labor and marriage markets that summarize our story:

- ① **Spatial Structural Changes (SSC)** $\left\{ \begin{array}{l} \text{Educational shifter (more educated women than men)} \\ \text{Sectoral shifter (women sort to the service sector)} \\ \text{Spatial sectoral shifter (more so in developed regions)} \end{array} \right\}$
- ② **Persistent social norms:** Females marry up, males marry down;

⇒ **Local marriage mismatch:** $\left\{ \begin{array}{l} \text{Low-skill single men in undeveloped regions;} \\ \text{High-skill single women in developed regions.} \end{array} \right\}$

This Paper (Empirical)

- "Squeeze effect" and mismatch in local marriage markets:



[With Spatial Structural Change]

This Paper (Quantitative)

- ▶ Build a prefecture-level spatial equilibrium model:
 - * multi-sector and multi-skill production;
 - * migration across prefectures;
 - * local marriage market.
- ▶ Embed a marriage matching model à la Choo & Siow (2006).
- ▶ Parameterize the model to match the Chinese economy in 2015.
- ▶ Decompose & quantify the sources of observed spatial marriage patterns.
- ▶ Counterfactual policies.

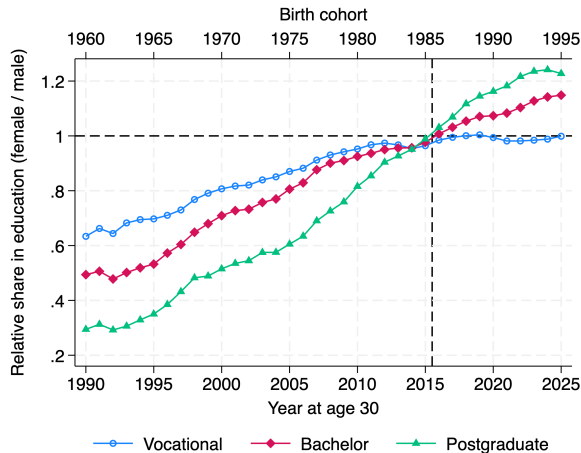
Preview of Results

- ▶ Gender-specific spatial structural changes matter: If we remove gender specificity:
 - * National singles rate ↓ 30% for females; ↓ 12% for males.
 - * Driven by high-skilled (low-skilled) females (males) in more developed (less developed) regions.
- ▶ We then decompose the SSCs into three parts:
 - * **Educational shifter** explains one-third of this decline.
 - * **Spatial sectoral shifter** explains the remaining two-thirds.
 - * Overall sectoral change alone plays a minimal role.
- ▶ We project that if the SSC continues to 2030:
 - * Spatial mismatch ↑ in China.
 - * Singles rate ↑ 60% for females and 20% for males.
- ▶ Marriage subsidies have a very limited policy effect, sadly.

Empirical Patterns

Dramatic Gender-specific Structural Changes

- **Gender educational gap** for females narrows and reverses over time:



*For birth cohorts after 2005, undergraduate admission female/male has been over 2:1.

Dramatic Gender-specific Structural Changes

- **Gender employment gap** in the service sector decreases for females, particularly high-skill:

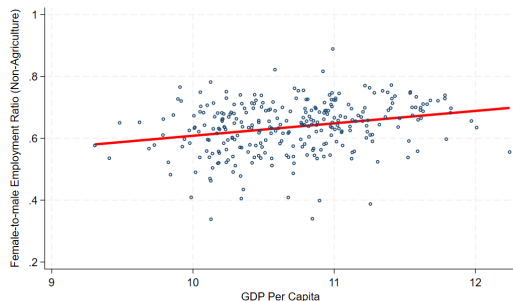
Education	Sector	2000	2005	2010	2015
College and Above	Agriculture	-45.0%	-32.3%	-16.2%	-13.2%
	Manufacturing	-34.6%	-26.9%	-23.3%	-22.7%
	Service	-16.6%	-1.4%	+12.0%	+21.2%
High School	Agriculture	-39.5%	-38.6%	-24.3%	-17.5%
	Manufacturing	-22.1%	-28.9%	-29.8%	-33.1%
	Service	-0.4%	-3.4%	+1.1%	+4.1%
Middle School and Below	Agriculture	+14.9%	+17.8%	+19.0%	+18.6%
	Manufacturing	-18.6%	-18.0%	-17.7%	-24.1%
	Service	-14.6%	-11.2%	+4.7%	+9.3%

- **Gender wage gap** in the service sector also narrows more for college females.

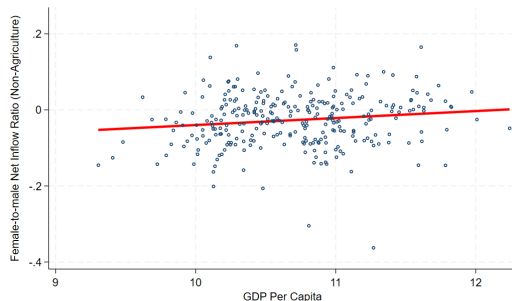
Dramatic Gender-specific Structural Changes

► Gender spatial employment gap (female - male):

Females are more likely to work in non-agricultural sectors in developed regions.



(a) Overall Employment (female - male)



(b) Net Migration Inflow (female - male)

Persistent Social Norm in Marriage Matching

- ▶ Married males have **higher education level** than their unmarried counterparts
- ▶ Married females have **lower education level** than their unmarried counterparts

Table Relative Socioeconomic Status Gap of [Married - Never-married]

Census Year	2000		2005		2010		2015	
	Male	Female	Male	Female	Male	Female	Male	Female
College Degree	+0.05	-0.09	+0.04	-0.12	+0.04	-0.08	+0.06	-0.14
Education Year	+2.41	-0.56	+2.13	-0.62	+1.33	-0.69	+1.49	-0.66

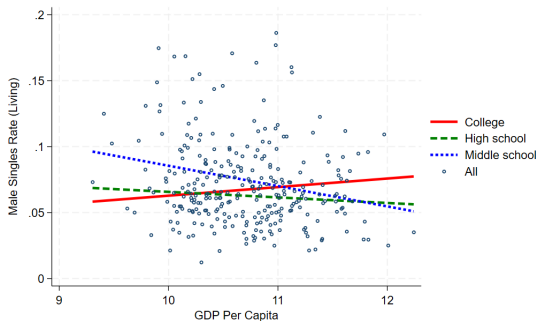
Preference

Persistent Social Norm in Marriage - homogamy/hypergamy

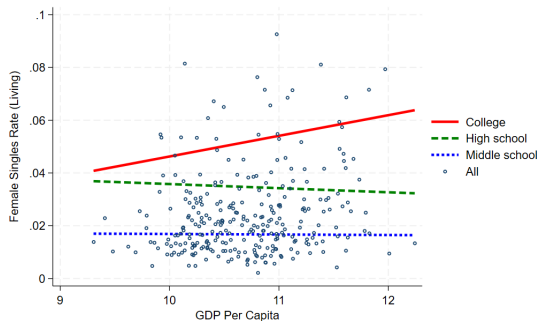
Table Relative Socioeconomic Status of Married Couples

Census Year	2000	2005	2010	2015
<i>Education Year</i>				
Females marry up	38.61%	37.90%	30.01%	28.33%
Females marry down	9.29%	9.67%	8.96%	9.77%
Equal	52.10%	53.23%	61.04%	61.90%

Spatial Distribution of Singlehood



(a) Male



(b) Female

Figure GDP and Singles Rate of Age 30-45 in 2015 (City-level, living pop)

Service Share

A Spatial Equilibrium Model with Marriage Matching

Overview of the Model

A quantitative spatial equilibrium migration model (Eaton & Kortum, 2002; Tombe & Zhu, 2019; Fang et al., 2022)

→ Embedded with a marriage matching model (Choo & Siow, 2006)

► **A set of prefectures** indexed by $i = 1, \dots, N$, each with three sectors:

- * [manufacturing , service] $\Leftarrow$ combine different skilled labor
- * [agriculture] $\Leftarrow$ indifferent labor

► **A measure of workers** H_i : endowed with gender, skill, hometown.

- * Start as single;
- * Migrate, work;
- * Then participate in **local marriage market**.

Location Preference and Migration

- ▶ For an individual o of gender $g = \{male, female\}$ and edu/skill $e = \{high, med, low\}$, migrate from city i to city j and work in sector k :

$$U_{i,jk}^o = \underbrace{\bar{V}_{jk}^{ge}}_{\text{expected payoff in } jk} \cdot \underbrace{\frac{1}{\tau_{i,jk}^{ge}}}_{\text{allocation cost}} \cdot \underbrace{z_{i,jk}^o}_{\text{location pref. shock}}$$

- ▶ Compound allocation cost $\tau_{i,jk}^{ge}$: flexible type- & flow-specific.
- ▶ Gravity equation for the migration flow $\pi_{i,jk}^{ge}$.

Workers: Utility of Married Individuals

- ▶ A worker o in destination jk , married to e' -type spouse to form a household ω , has a log-linear utility:

$$V_{jk}^{\omega,o} = \ln \left[\underbrace{\left(\frac{c_{jk}^{\omega}}{\beta} \right)^{\beta}}_{\text{HH final goods consumption}} \cdot \underbrace{\left(\frac{h_{jk}^{\omega}}{1-\beta} \right)^{1-\beta}}_{\text{HH housing consumption}} \right] \underbrace{\bigg/ (1+\chi)}_{\text{economy of scale}} + \underbrace{m_j^{\omega,o}(e')}_{\text{marital payoff}}$$

- ▶ Single workers enjoy own consumption, housing, and normalized value of single $m_j^o(\emptyset)$.

Marriage Market

- ▶ In destination city j , there is a local marriage market (**across sectors**).
- ▶ Following [Choo & Siow \(2006\)](#), assume transferable utility (TU), individual o 's marital payoff from marrying a type- e' spouse:

$$m_j^{\omega,o}(e') = \underbrace{\tilde{\mu}_j^{ge}}_{\text{value of not single in } j} + \underbrace{\mu^{ge}(e')}_{\text{deterministic partner pref.}} + \underbrace{\delta_j^{ge}(e')}_{\text{eq'm marital transfer in } j} + \underbrace{\varepsilon_j^o(e')}_{\text{idiosyncratic partner pref.}}$$

- * $\mu^{ge}(e')$ is deterministic marital return (or love) relative to being single;
- * $\delta_j^{ge}(e')$ is equilibrium transfer within couple, and $\delta_j^{ge}(e') = -\delta_j^{g'e'}(e)$.

- ▶ Identification comes from variations in marriage matching [equilibria across locations](#).

Production

- Sector- k (manufacturing/service) produce single final good Y by aggregating skills (e) in CES:

$$Y_{jk} = \left[\sum_e (A_{jk}^e H_{jk}^e)^{\frac{\sigma_k - 1}{\sigma_k}} \right]^{\frac{\sigma_k}{\sigma_k - 1}}$$

- * A_{jk}^e is city-level efficiencies of using e -skilled labor in sector k ,
- * *Effective* labor supply H_{jk}^e accounts for gender wage/participation gaps.
- * In rural regions, production is simply $Y_{jr} = A_{jr} H_{jr}$, regardless of skill level.

Housing Market

- ▶ Essentially a congestion mechanism.
- ▶ Housing demand equals the supply of floor space allocated to residential use S_{ju} :

$$\underbrace{\phi_j L_{ju}}_{\text{Housing supply}} = S_{ju} = \underbrace{\mathbb{E}[h_{ju}] H_{ju}}_{\text{Housing demand}}$$

- * where highly-regulated construction sector uses geographic land L_{ju} and a regulated density of development ϕ_j to produce floor space S_{ju} .

Model Solution and Estimation

Data and Parameters

- ▶ Census 2015:
 - * Population flow; marriage matching; individual characteristics
- ▶ City Statistical Yearbooks 2015: for each prefecture.
 - * Wages; land supply; GDP by sector.
- ▶ China Real Estate Information (www.crei.cn):
 - * Housing price.
 - * (Run by the National Development and Reform Commission)

- ◆ Calibrated parameters from the literature/data [Detail](#)

Solve and Estimate the Spatial Equilibrium

► **Inner loop:** Marriage market equilibrium.

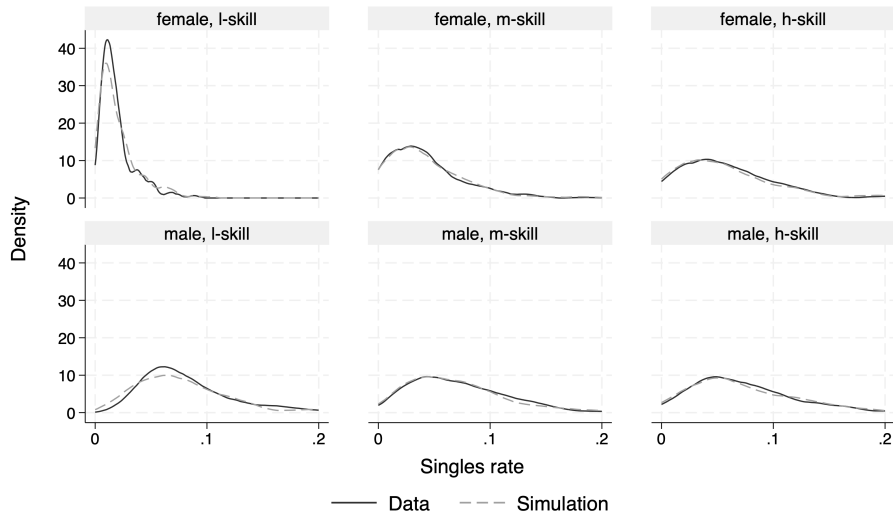
- We observe local marriage market equilibria across multiple locations. [Detail](#)

► **Outer loop:** Spatial general equilibrium.

- Invert the spatial model to calibrate: [Detail](#)

- * Productivities A_{jk}^e
- * Floor space construction intensity ϕ_j
- * Compound allocation costs $\tau_{i,jk}^{ge}$

Model Fit: Spatial-Distribution of Singles Rate



Estimated Model Parameters - Marital Preference/Social Norms

- For each search target (e'), the deterministic marital return $\mu^{ge}(e')$:

Male type	Wife type			Female type	Husband type		
	l-skill	m-skill	h-skill		l-skill	m-skill	h-skill
l-skill	3.312	-0.672	-3.589	l-skill	5.253	2.373	-0.277
m-skill	2.615	2.654	0.106	m-skill	2.707	3.850	2.280
h-skill	0.651	1.770	3.175	h-skill	-0.101	1.411	3.794

- For females, they do not like to marry down
- For males, they do not like to marry up

Estimated Model Parameters - Spatial/Sector Allocation Cost

- After recovering the allocation/migration cost τ , we further decompose it:

$$\ln(\tau_{i,jk}^{ge}) = G(d_{i,j}) + \overline{\tau}_k^{ge} + \varepsilon_{i,jk}^{ge}$$

- * $G(d_{i,j})$: a third-order polynomial for distance;
- * $\overline{\tau}_k^{ge}$: **average sectoral** allocation cost (no location variation)
- * $\varepsilon_{i,jk}^{ge}$: **sector-location specific** allocation cost

Estimated Model Parameters - Sector Allocation Cost $\bar{\tau}_k^{ge}$

- ▶ We normalize the allocation cost for high-skilled females in the service sector to zero
- ▶ Cost is low in the service sector for high-skilled workers regardless of gender
- ▶ Cost is low in the service sector for females regardless of skill

Table Relative Sectoral Allocation Costs by Gender and Skill

$\bar{\tau}_k^{ge}$	Male			Female		
	l-skill	m-skill	h-skill	l-skill	m-skill	h-skill
Agriculture	0.398	0.821	1.581	0.319	0.857	1.785
Manufacturing	0.126	0.159	0.215	0.178	0.249	0.353
Service	0.290	0.189	0.080	0.224	0.097	0

Estimated Model Parameters - Average Location Cost $\varepsilon_{i,jk}^{ge}$

- ▶ We average $\varepsilon_{i,jk}^{ge}$ over sectors and normalize the allocation cost for high-skilled females in the most developed cities to zero
- ▶ Cost is low in developed cities regardless of skill and gender
- ▶ Especially for high-skilled females

Table Relative Spatial Sectoral Allocation Costs by Gender and Skill

Average $\varepsilon_{i,jk}^{ge}$	Male			Female		
	l-skill	m-skill	h-skill	l-skill	m-skill	h-skill
Least Developed	0.463	0.488	0.592	0.473	0.564	0.629
Second Quartile	0.555	0.600	0.594	0.554	0.570	0.610
Third Quartile	0.448	0.445	0.417	0.439	0.425	0.399
Most Developed	0.171	0.104	0.034	0.171	0.079	0

Quantitative Analysis

Quantitative Analysis

- ▶ Decomposition of the spatial structural changes (SSC) effect SSC Effect
- ▶ What is the future of China in 2030? China 2030
- ▶ Universal marriage subsidies Marriage Subsidies

Conclusion

Quantitative Analysis I: SSC Effect

- ▶ In the first quantitative analysis, we try to evaluate the impact of the spatial structural changes on singles rate
- ▶ We consider three adjustments for males and females:
 - * (1) Equalizing gender education levels
 - * (2) Equalizing gender sectoral allocation costs $\bar{\tau}_k^{ge}$
 - * (3) Equalizing gender spatial-sectoral allocation costs $\epsilon_{i,jk}^{ge}$
- ▶ We first equalize (1), (2), (3) at the same time to erase all gender-specific SSCs
- ▶ Then, we equalize them one by one to implement a decomposition

Quantitative Analysis I: SSC Effect

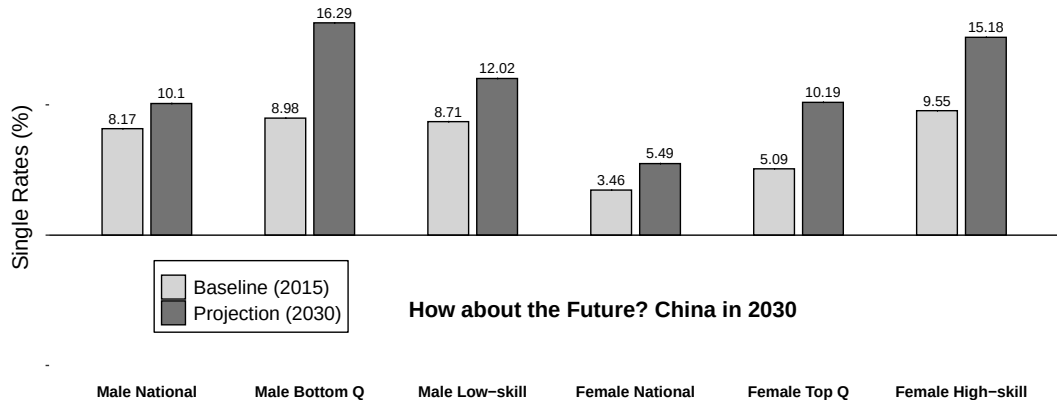
Table The Effects of Gender-specific Spatial Structural Changes on Singles Rate

National & Regional Singles Rate	Male			Female		
	National	Least Dev.	Low Skill	National	Most Dev.	High Skill
Panel A: Singles Rate and Percentage Changes						
Baseline	8.17%	8.98%	8.71%	3.46%	5.09%	9.55%
No GS-SSCs	7.21%	8.03%	7.63%	2.45%	3.11%	4.46%
% Changes	-11.75%	-10.58%	-12.40%	-29.19%	-38.90%	-53.30%
Panel B: Decomposition of the Percentage Changes						
National Educational	32.29%			31.68%		
National Sectoral	-1.04%			0.00%		
Spatial Sectoral	68.75%			68.32%		

Quantitative Analysis II: China in 2030

- ▶ In the second analysis, we predict what will happen if the SSC continues to 2030:
 - * Gender college education rate is determined by gender college enrollment rate of cohort aged 20 in Census 2020;
 - * Gender-specific sectoral shifts are projected linearly using trend from 2000 to 2015;
 - * Gender gap in spatial sectoral allocation costs are doubled.

Quantitative Analysis II: China in 2030

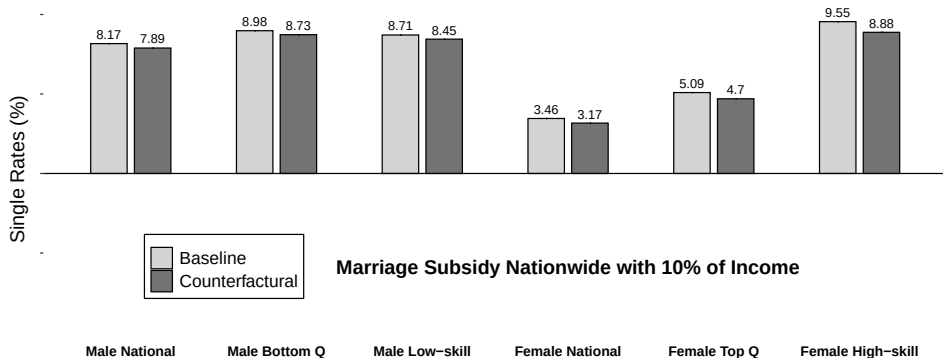


Quantitative Analysis III: Marriage Subsidies

- ▶ We inject a universal marriage subsidy of 10% lifetime family income (not from govt. budget).
 - * Receives as long as getting married.
- ▶ This would cost about 3.5% of the national GDP per year.
 - ⇒ Similar to that in Nordic countries combining all family policies.
- ▶ Much larger than most of the pure marriage policies: e.g.
 - * Busan, Korea: \$15,000
 - * Guangzhou, China: \$8,000

Quantitative Analysis III: Marriage Subsidies

- ▶ The policy effect is very small, despite large fiscal burden.
- ▶ A pure monetary reward cannot alter the *fundamental spatial mismatch* from SSCs.

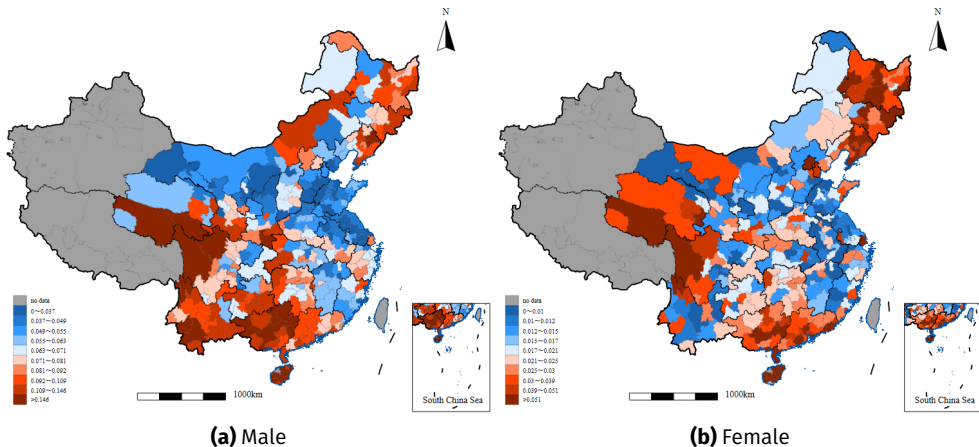


Conclusion

- ▶ This paper develops a spatial GE migration model incorporated with local marriage matching.
- ▶ We closely match the spatial disparity of marriage outcomes by gender and skill.
 - * A race between the persistent social norm and the dramatic spatial structural changes.
- ▶ Using the spatial GE model, we find that
 - * SSC accounts for 30 (12) percent of the singles rate for females (males) in China.
 - * Reversed gender educational gap and gender-specific spatial sectoral allocation are the most important contributors.
 - * Marriage rate in China is likely to continue to drop amid current trend.
 - * Marriage subsidy is costly and relatively ineffective.

Appendix

Marriage and Spatial Sorting: Motivation [Back](#)

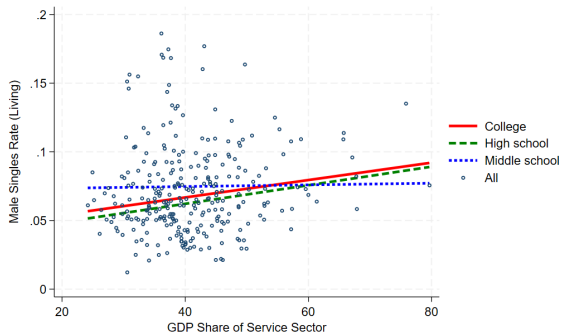


(a) Male **(b) Female**
Figure Prefecture-level Singles Rate of People over 35 in China

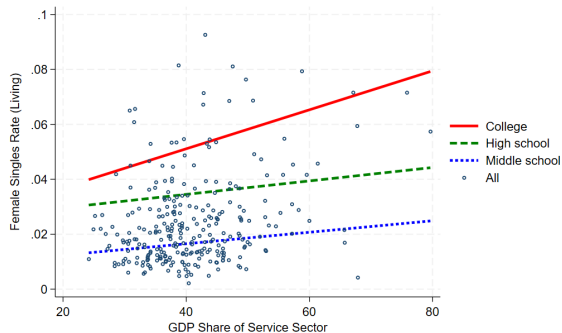
- ▶ Marriage is still important in China
- ▶ Young Chinese people still want to get married

% don't want to get married:		
(Age 18-45)		
	Male	Female
CFPS 2018	2.54	3.41
CFPS 2020	2.41	2.88

Spatial Distribution of Singlehood - Service Sector Share Back



(a) Male



(b) Female

Figure Employment Share in Service and Singles Rate of Age 30-45 (City-level, living pop)

Location Preference and Migration

- ▶ After location pref. shock realized, individual chooses city-sector jk to live and work,
 - * given $\bar{V}_{jk}^{ge}$ (wage, housing price, marriage market return) and migration cost $\tau_{i,jk}^{ge}$.
- ▶ Denote $\pi_{i,jk}^{ge}$ as the share of workers of gender g and skill e , from home i , migrating to jk .
- ▶ Gravity equation for the migration flow:

$$\pi_{i,jk}^{ge} = \frac{(\tau_{i,jk}^{ge})^{-\sigma_z} (\bar{V}_{jk}^{ge})^{\sigma_z}}{\sum_{j'k'} (\tau_{i,j'k'}^{ge})^{-\sigma_z} (\bar{V}_{j'k'}^{ge})^{\sigma_z}} = \frac{\Phi_{i,jk}^{ge}}{\Phi_i^{ge}}$$

Spatial General Equilibrium

► A Spatial General Equilibrium is defined by

- * a set of initial conditions and exogenous technology $\{\tau_{i,jk}^{ge}, A_j^e, \phi_j, L_j, H_i^{ge}\}$,
- * endogenous prices $\{w_{jk}^e, q_{ju}\}$ and marital transfer $\{\delta_j^{ge}(e')\}$,
- * endogenous quantities $\{Y_{jk}, H_{jk}^{ge}, c_{jk}^{ge}, h_{jk}^{ge}, p_{jk}^{ge}, S_{ju}\}$,
- * and distributions $\{\pi_{i,jk}^{ge}\}$,

such that, under perfect foresight in labor and marriage market, followings are achieved:

- (1). [Worker optimization];
- (2). [Firm optimization];
- (3). [Goods and land market clearing];
- (4). [Marriage market stable matching].

Calibrated parameters from the literature/data [Back](#)

Parameter	Description	Value	Source
<i>From literature</i>			
β	share of consumption in utility	0.77	Urban Household Survey
ξ	relative cost of rural housing	0.34	Census
σ_k	elasticity of skill substitution	4.0	Bils et al.(2024)
σ_z	migration elasticity	1.5	Tombe & Zhu (2019)
σ_ε	dispersion of marriage pref.	0.734	Marriage mkt. estimation
<i>From microdata</i>			
H_i^{ge}, H_{jk}^{ge}	Workers by type by location		Census
$\pi_{i,jk}^{ge}$	Migration flows between <i>in</i> and <i>jk</i>		Census
w_{jk}^e	Wage of skill skill <i>e</i> in city <i>j</i> sector <i>k</i>		Stat. Yearbook + Census
r_{jk}^e	Gender wage/participation gap of <i>e</i> in <i>jk</i>		Census + CFPS
q_{jk}	Unit rent in city <i>j</i> region <i>k</i>		Housing prices

Solve the Inner Loop - Marriage Market Equilibrium Back

- ▶ We observe local marriage market equilibria across multiple locations.
- ▶ Use the property of the logit-type matching shares and log-linearization:

$$\begin{aligned}\ln[P_j^{ge}(e')] - \ln[P_j^{ge}(\emptyset)] \\ = \frac{1}{\sigma_\varepsilon} \left[\tilde{\mu}_j^{ge} + \mu^{ge}(e') + \delta_j^{ge}(e') + \ln \left(\frac{W_j^{ge} + W_j^{g'e'}}{W_j^{ge}(1+\chi)} \right) \right]\end{aligned}$$

and symmetry constraint for marital transfers $\delta_j^{ge}(e') = -\delta_j^{g'e'}(e)$.

- ▶ Values on the LHS is observed
- ▶ It is over-identified $\Rightarrow$ estimate the parameters using GMM
 - * $\tilde{\mu}_j^{ge}$ can be identified by the overall singles rate in each location j
 - * $\mu^{ge}(e')$ can be identified by match rate of each pair (e, e') (no location variation)
 - * $\delta_j^{ge}(e')$ can be identified by match rate of each pair across locations

Solve and Estimate the Spatial Equilibrium (Outer Loop) [Back](#)

- ▶ With data of population distributions, migration flows, prices, and expected marital payoff in each city, we can invert the model to calibrate:
 - Productivities A_{jk}^e
 - * Under profit maximization and zero profits, inferred from employment and wages. [Detail](#)
 - Floor space construction intensity ϕ_j
 - * Worker's FOC for housing gives required floor space, then back out construction intensity. [Detail](#)
 - Migration costs $\tau_{i,jk}^{ge}$
 - * With all others known, calculated for each type and city-pair using the Gravity Equation. [Detail](#)

Solve the spatial equilibrium: Productivities Back

- ▶ From profit maximization and zero profits, we can infer urban sectoral productivity from the data on employment and wages for $k = \{m, s\}$.
- ▶ First, we solve for productivity A_{jk}^h as a function of A_{jk}^l using the first order conditions $A_{jk}^h = A_{jk}^l (H_{jk}^h / H_{jk}^l)^{1/(\sigma_k-1)} (w_{jk}^h / w_{jk}^l)^{\sigma_k/(\sigma_k-1)}$.
- ▶ Plugging A_{jk}^h into the definition of Y_{jk} , we have:

$$Y_{jk} = A_{jk}^l H_{jk}^l \left[\frac{w_{jk}^h H_{jk}^h + w_{jk}^l H_{jk}^l}{w_{jk}^l H_{jk}^l} \right]^{\frac{\sigma_k}{\sigma_k-1}} \equiv A_{jk}^l H_{jk}^l (\Xi_{jk}^l)^{-\frac{\sigma_k}{\sigma_k-1}}$$

where $\Xi_{jk}^l = \frac{w_{jk}^l H_{jk}^l}{w_{jk}^h H_{jk}^h + w_{jk}^l H_{jk}^l}$ is the share of labor income distributed to low skill workers.

- ▶ We also assume that agricultural productivity equals agricultural wages $A_{jr}^e = w_{jr}$, for both $e = \{h, m, l\}$. Intuitively, higher wages or skill shares require higher skill s productivity at equilibrium for urban sectors.
- ▶ We can then calculate the productivities for both skill types as follows:

$$A_{jk}^l = w_{jk}^l (\Xi_{jk}^l)^{\frac{1}{\sigma_k-1}}, \quad A_{jk}^h = w_{jk}^h (1 - \Xi_{jk}^l)^{\frac{1}{\sigma_k-1}}.$$

Solve the spatial equilibrium: Land market clearing [Back](#)

- From workers' first-order conditions for residential floor space and the summation of all workers residing in each prefecture and region jk , we can calculate both urban and rural floor space:

$$S_{ju} = \frac{1-\beta}{q_{ju}} \sum_k [w_{jk}^l H_{jk}^l + w_{jk}^m H_{jk}^m + w_{jk}^h H_{jk}^h], \quad S_{jr} = \frac{1-\beta}{q_{jr}} [w_{jr} H_{jr}]$$

- We can then back out the implied construction intensity $\phi_j = S_{ju}/L_j$.

Solve the spatial equilibrium: Migration costs Back

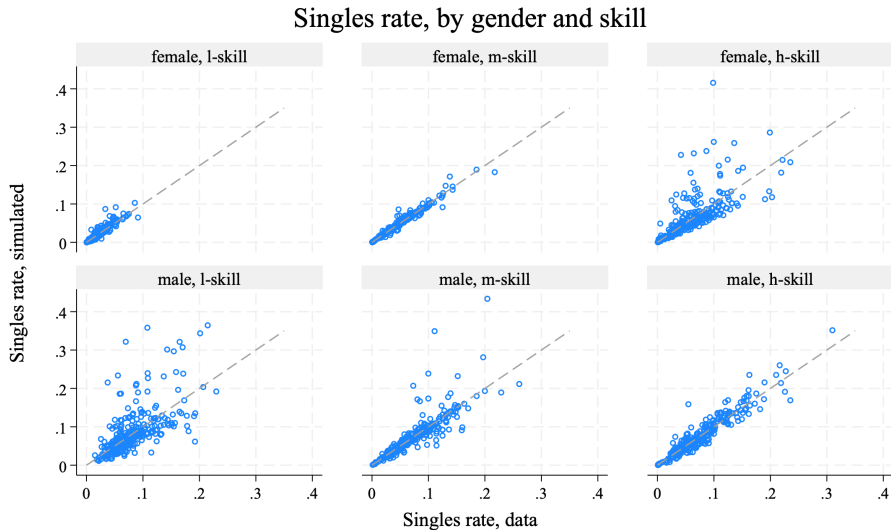
- ▶ We first to compute the prefecture-level equally-divided rent income for residents $\frac{q_i S_i}{H_i}$ from the residential floor space S_i calculated above, to which we can add observed wages to determine incomes of workers of skill e moving from in to jk : $inc_{i,jk}^e = w_{jk}^e + \frac{q_{jn} S_{jn}^R}{H_i^R}$.
- ▶ Second, we need to calculate the workers' marriage utility $m_{i,jk}^{ge}$.
- ▶ Then, we can calculate all migration costs between all prefecture pairs from the gravity equations.
- ▶ We assume the iceberg migration cost for staying in one's original prefecture is $\tau_{i,in}^{ge} = 1$.
- ▶ With q_i , $inc_{i,jk}^e$, $m_{i,jk}^e = E(m_{i,jk}^{ge})$, and $\pi_{i,jk}^e = \sum_{ge} \pi_{i,jk}^{ge}$ in hand, along with the gravity equation:

$$\Phi_i^e = \sum_{jk} (\tau_{i,jk}^{ge} q_{jk}^{1-\beta})^{-\sigma_z} (inc_{i,jk}^e m_{i,jk}^e)^{\sigma_z} = \frac{(q_{jk}^{1-\beta})^{-\sigma_z} (inc_{i,in}^e m_{i,jk}^e)^{\sigma_z}}{\pi_{i,in}^e}$$

- ▶ By inserting Φ_i^e into the original gravity equation, we have:

$$\tau_{i,jk}^e = \frac{ic_{i,jk}^e m_{i,jk}^e}{q_{jk}^{1-\beta} (\pi_{i,jk}^e \Phi_i^e)^{1/\sigma_z}}, \text{ for } in \neq jk$$

Model Fit: City-Level Singles Rate by Gender and Skill



Related Literature

► **Gender-skewed migration and sorting:**

- * Young women outnumber men in urban area (Edlund, 2005; Ong, Yang & Zhang, 2020; Koh, Li, Wu, Yi & Zhang, 2024); Migrate for marriage (Dupuy, 2021; Zhou, 2021; Xiong, 2023); Gender wage gap across regions (Bacolod, 2017).

► **Demography and marriage matching (Weiss, 1997; Choo and Siow, 2006):**

- * Assortative mating (Hitsch et al., 2010; Abramitzky et al., 2011; Dupuy and Galichon, 2014), particularly strong in education (Eika et al., 2019; Ashraf et al., 2020).
- * Women prefer hypergamy (Fisman et al., 2006; Schwartz and Mare, 2005; Bertrand et al., 2015)

⇒ We bridge the two and introduce the local marriage market in a quantitative spatial model.

⇒ Highlight the gender/skill pattern of migration in shaping marriage matching.

Related Literature

- ▶ **Structural change and the rise of the service sector** (especially in China, see Chen, Pei, Song & Zilibotti, 2023):
 - * Gender gaps (Ngai & Petrongolo, 2017; Autor, Dorn & Hanson, 2019); Female relative advantage in higher-skilled jobs (Cortes, Jaimovich & Siu, 2018); Overall trends of marriage (Greenwood, Guner & Vandenbroucke, 2017) and fertility (Doepke, Hannusch, Kindermann & Tertilt, 2023); Structural change (Kuhn, Manovskii & Qiu, 2024).

- ⇒ SC is gender-biased and spatial, sorting high-skilled females to developed regions, leading to spatial marriage mismatch.

- ▶ **Quantitative spatial models** (Redding & Rossi-Hansberg, 2017), and recent "spatial marriage" literature:
 - * Spatial distribution of skill (Fan & Zou, 2022; Alonzo, 2022); Assortative mating (Mao & Wen, 2024); Segregation and inequality (Alonzo, Guner & Luccioletti, 2023); Gender inequality in welfare (Deng, 2023).

- ⇒ We examine how spatial marital mismatch affects local and aggregate marriage trend.